
Software Requirements Specification

for

QuickHire

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for **QuickHire**, a web-based job recruitment and hiring platform, **Release 1.0**. QuickHire is designed to help job seekers easily find employment opportunities and participate in job interviews through an online system.

The scope of this SRS covers the core features of the QuickHire website, including user registration and login, job searching and browsing, and interview scheduling and conducting through the platform. The system focuses on supporting job seekers in finding suitable jobs and enabling employers to interview applicants efficiently.

This SRS does not cover advanced human resource management functions such as payroll processing, employee performance tracking, or post-employment management. It is limited to the job discovery, application, and interview processes that occur prior to hiring.

1.2 Document Conventions

This SRS for **QuickHire** follows standard software documentation guidelines (IEEE SRS). Clear headings and numbered sections are used for easy reading. Requirement statements use “**shall**” for mandatory features, “**should**” for recommended features, and “**may**” for optional features.

Each requirement has its **own priority level** (High, Medium, or Low), and priorities are **not inherited** from higher-level requirements. A consistent font style and size are used throughout the document, with **bold text** applied to section titles for emphasis.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for **developers, project managers, testers, and users** involved in the development and use of the **QuickHire** website. Developers and testers will use this document to understand system requirements and expected behavior, while project managers will use it for planning and tracking progress. Users may refer to this document to understand the system’s features and limitations.

The document is organized into sections that describe the system overview, functional requirements, non-functional requirements, and system constraints. Readers are encouraged to begin with the **introduction and overall description** to understand the purpose of QuickHire, then proceed to the **specific requirements sections** most relevant to their role.

1.4 Product Scope

QuickHire is a web-based job recruitment platform for IT workers like developers, data analysts, system analysts and other IT related jobs designed to help job seekers quickly find suitable employment and participate in interviews online. The goals of QuickHire are to reduce the time and effort needed to find a job, improve communication between employers and applicants, and support businesses in hiring qualified candidates efficiently. The software aligns with business goals by helping organizations reach a wider pool of applicants, streamline recruitment processes, and make faster hiring decisions.

1.5 References

Abardo, L. S., et al. (2025). SCAMBA: A Deep Learning-Based Android Spam Classifier for Monolingual and Code-Switched SMS in the Philippines.

IEEE Computer Society. (1998). IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

QuickHire is a new, self-contained web-based job recruitment platform developed to address common problems in traditional job searching and hiring processes in IT fields. It is not a replacement for an existing system nor part of a previous product family, but a standalone solution designed specifically to connect job seekers and employers in one online environment.

QuickHire interacts with external systems such as web browsers, internet services, and optional third-party communication tools (e.g., email or video conferencing) to support notifications and interviews. All core functionalities are handled within the QuickHire system itself.

2.2 Product Functions

Major Product Functions

At a high level, **QuickHire** provides the following major functions:

- User registration and login for job seekers and employers
- Job searching and filtering by job seekers
- Job posting and listing for employers
- Messaging for job seeker and employers
- Interview scheduling and notification
- Online interview support (e.g., chat or video integration)
- Profile management for users
- System notifications (application status, interview updates)

These functions work together to support the complete job search and hiring process, from finding job opportunities to conducting interviews.

2.3 User Classes and Characteristics

User Classes and Characteristics

The QuickHire system is expected to be used by the following user classes:

- **Job Seekers:** Job seekers are the primary users of the system. They use the platform frequently to create profiles, search for jobs, manage profiles, report users, post feedback and attend direct interviews. They typically have basic to moderate technical skills and limited system privileges.
- **Employers / Recruiters:** Employers use the system to post feedback and conduct interviews. They are regular users with higher privileges than job seekers, including job management and applicant evaluation features.

Job seekers and employers are the **most important user classes**, as the system is primarily designed to support their needs.

2.4 Operating Environment

QuickHire is a **web-based application** that runs on standard computing devices such as desktop computers, laptops, tablets, and smartphones. Users can access the system through modern web browsers including **Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari**.

The system operates on common operating systems such as **Windows, macOS, Android, and iOS**. On the server side, QuickHire may run on a cloud-based hosting environment with a standard web server, database server, and internet connectivity. The software is designed to coexist smoothly with other applications running on the user's device without requiring special hardware or software installations.

2.5 Design and Implementation Constraints

The development of **QuickHire** is subject to several constraints. The system must comply with applicable **data privacy and security regulations** to protect user information. As a web-based platform, it is limited by **internet connectivity** and standard browser capabilities.

The software will be developed using **web technologies** and must be compatible with commonly used browsers and operating systems. Integration with external services such as **email or video conferencing tools** may limit design options based on their APIs and protocols. Security measures such as **user authentication and data encryption** must be implemented according to accepted industry standards.

In addition, the system should follow standard **coding and design conventions** to ensure maintainability, especially if future updates or maintenance are handled by a different development team.

2.6 User Documentation

The following user documentation will be provided with **QuickHire**:

- **User Manual:** A step-by-step guide for job seekers and employers on how to use the system.
- **Online Help:** Context-sensitive help available within the website for quick guidance.

Documentation will be delivered in **digital formats** such as **PDF, web pages, or embedded help modules** within the QuickHire platform. The content will follow standard usability and accessibility guidelines to ensure clarity and ease of use.

2.7 Assumptions and Dependencies

The following assumptions and dependencies have been identified for **QuickHire**:

- Users have access to a stable **internet connection** and a modern web browser.
- The system will rely on **third-party services** such as email providers and video conferencing tools for notifications and online interviews.
- Job seekers and employers are assumed to have basic **computer and internet skills** to navigate the platform.
- The system depends on standard **cloud hosting services** and database platforms for storage and scalability.
- Any changes in browser compatibility, third-party APIs, or internet regulations may affect system functionality.

These assumptions are critical for the system to operate as intended, and any changes may require updates to the requirements or design.

3. External Interface Requirements

3.1 User Interfaces

QuickHire provides a **web-based graphical user interface (GUI)** that allows job seekers and employers to interact with the system through standard web browsers. The interface is designed to be **simple, responsive, and easy to navigate**, supporting users with basic to moderate technical skills.

General Interface Characteristics

The QuickHire interface shall follow modern web design standards and maintain consistency across pages. The system shall use a **dashboard-based layout with a left-side navigation menu and a central content display area**.

The interface includes the following visual components observed in the system:

3.1.1 Navigation Sidebar

The system shall provide a **persistent sidebar navigation menu** located on the left side of the screen. The sidebar contains the following options:

- **Find Employer:** Allows job seekers to browse and search for available employers or job opportunities, helping them start the job matching and application process.
- **Home:** Displays the main dashboard where users can view their profile summary, notifications (such as interview updates), and overall activity within the system.
- **Edit Profile:** Enables users to update and manage their personal and professional information, including skills, resume, job preferences, and other profile details used for job matching.
- **Logout:** Securely exits the user from the QuickHire system to protect account information and prevent unauthorized access.

The sidebar also displays the **user profile information**, including:

- **Profile picture:** Displays the user's photo or avatar, helping employers visually identify the job seeker and personalize the profile.
- **User name:** Shows the full name or username of the account owner, used to identify the user within the system.
- **Country:** Indicates the user's current location or country of residence, which helps employers consider location-based opportunities or remote work compatibility.
- **Hourly rate:** Represents the expected pay per hour set by the job seeker, allowing employers to assess if the candidate fits their budget.

This navigation menu allows users to quickly switch between system features.

3.1.2 Dashboard Interface

The QuickHire dashboard provides an overview of the user's profile and activity. The dashboard shall display:

- **Welcome message for the user:** Displays a greeting message upon login to personalize the user experience and confirm successful access to the dashboard.
- **Profile status indicating visibility to employers:** Shows whether the user's profile is visible or hidden to employers, helping the user understand if they can currently be discovered for job opportunities.
- **Notification alerts (e.g., incoming interview call):** Provides real-time updates and important messages such as interview invitations, application status changes, or system alerts to keep the user informed.
- Profile summary containing:
 - **Job role:** Specifies the position or type of job the user is applying for (e.g., developer, data analyst), helping employers understand the user's career focus.
 - **Employment type:** Indicates the preferred work arrangement, such as full-time, part-time, freelance, or contract.
 - **Hourly rate:** Shows the user's expected compensation per hour, allowing employers to evaluate budget compatibility.
 - **Availability:** Describes the number of hours or schedule the user is available to work, helping employers plan job assignments.
 - **Language proficiency:** Indicates the user's level of fluency in languages (e.g., English), which is important for communication in the workplace.
 - **Selected skills:** Lists the skills chosen by the user (e.g., programming, design, analytics), showcasing their qualifications and expertise.
 - **User overview or description:** Provides a short personal summary or introduction about the user, highlighting experience, strengths, and career goals.

Buttons available on the dashboard include:

- **Find Employer:** Allows the user to search and browse employers or job opportunities, helping them start the job application and matching process.
- **Edit Profile:** Enables the user to modify and update their personal, professional, and job-related information to improve their profile visibility and accuracy.

These buttons allow users to begin the job matching process or update their profile information.

3.1.3 Profile Editing Interface

The system provides a **profile editing page** where job seekers can update their information. The page includes a structured form layout with labeled fields.

The following fields are available for editing:

- Profile photo upload
- Desired job role
- Hourly rate
- Available working hours per day
- Country
- Employment type
- English proficiency level
- Educational background
- Portfolio or website link
- Age
- Gender
- Profile description

- Resume upload (PDF)

Users shall be able to upload or replace their resume file through the system.

3.1.4 Skills Selection Interface

The system includes a **skills selection panel** that allows users to select their professional skills. Skills are organized into categories such as:

- Artificial Intelligence
- Analytics
- API
- Backend
- Business
- Cloud
- Database
- Design
- Development

Users can search skills using a **search bar** and select multiple skills through checkboxes. Selected skills will be displayed in the **profile summary**.

3.1.5 Interface Standards Control

Interface Standards and Controls

The user interface shall follow these design standards:

- **Consistent color scheme and layout across all pages:** Ensures that all pages use the same design style, colors, and structure to provide a uniform and user-friendly experience.
- **Standard buttons such as Save Profile, Cancel, Edit, and Logout:** Uses familiar and clearly labeled buttons to help users easily perform common actions within the system.
- **Clear form labels and input fields:** Provides well-defined labels and fields so users can easily understand what information is required when filling out forms.
- **Confirmation messages after successful actions:** Displays feedback messages (e.g., "Profile updated successfully") to inform users that their actions have been completed.
- **Error messages displayed when invalid input is detected:** Alerts users when incorrect or incomplete information is entered, guiding them to fix the issue.
- **Responsive design compatible with desktops, tablets, and smartphones:** Ensures the system adapts to different screen sizes and devices for a smooth and accessible user experience.

Keyboard shortcuts and accessibility options may be supported to improve usability.

3.2 Hardware Interfaces

QuickHire is a **web-based application** that runs on common computing devices and does not require specialized hardware.

3.2.1 Supported Devices

The system shall support the following devices:

- Desktop computers
- Laptop computers
- Tablets
- Smartphones

3.2.2 Input Devices

Users interact with the system using standard input devices such as:

- Keyboard
- Mouse
- Touchscreen controls (for mobile devices)

3.2.3 Output Devices

The system displays information through:

- Computer monitors
- Mobile device screens

3.2.4 Hardware Interaction

QuickHire communicates with hardware indirectly through the **web browser and operating system**, which handle input and output operations. The system does not directly control hardware components. An internet-enabled device is required to access the platform.

3.3 Software Interfaces

QuickHire interacts with several external software systems and components.

3.3.1 Web Browsers

Users access QuickHire through modern web browsers including:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

These browsers support the rendering of HTML, CSS, and JavaScript required by the system.

3.3.2 Operating Systems

The system supports common operating systems including:

- Windows
- macOS
- Android
- iOS

3.3.3 Database Management System

QuickHire uses a database server to store and manage system data such as:

- **User profiles:** Stores personal and professional information of users, including their details, preferences, and account data.
- **Job applications:** Contains records of jobs that users have applied for, including application status and related details.
- **Skills and qualifications:** Stores information about the user's abilities, expertise, and credentials used for job matching.
- **Resume files:** Keeps uploaded resume documents (PDF) that employers can review for detailed qualifications.
- **Interview schedules:** Holds data about planned interviews, including date, time, and participants.
- **Notification records:** Stores system-generated messages such as alerts, updates, and reminders sent to users.

The database supports secure storage, retrieval, and updates of information.

3.3.4 Email Services

The system integrates with **external email service providers** to send notifications including:

- **Account registration confirmation:** An automated email sent to a user (job seeker or employer) after they successfully create an account on the QuickHire platform, confirming that their registration was completed.
- **Application status updates:** Notifications sent to job seekers informing them of changes in the status of their job applications for example, whether their application was reviewed, accepted, or rejected by an employer.
- **Interview invitations:** Notifications sent to job seekers when an employer schedules or requests an interview with them through the platform, which may include details about the scheduled online interview session.
- **System alerts:** General-purpose notifications generated by the QuickHire system to inform users of important events or actions that require their attention such as an incoming interview call, profile visibility changes, or other platform-related updates.

3.3.5 Video Communication Tools

QuickHire may integrate with **third-party video communication services** to support online interviews between employers and job seekers.

Data exchanged between QuickHire and external software includes:

- **User authentication data:** Information used to verify the identity of users (job seekers and employers) when logging into the QuickHire platform. This includes credentials such as usernames and passwords, which are protected through secure login authentication and password encryption via HTTPS.
- **Job application records:** Data stored in the QuickHire database that tracks the details of a job seeker's applications, including their application status and related profile information such as skills, qualifications, and uploaded resume files. These records are synchronized with the database in real time.
- **Notification messages:** System-generated communications delivered to users to keep them informed of relevant events on the platform, such as application status updates, interview invitations, and account confirmations. These may be delivered both within the platform (as dashboard alerts) and through external email service providers.
- **Interview schedules:** Records stored in the QuickHire database that contain the details of arranged interviews between employers and job seekers. These are synchronized in real time and may trigger notifications to both parties, and may also be coordinated through integrated third-party video communication services for conducting online interviews.

Communication between components is handled through secure **application programming interfaces (APIs)**.

3.4 Communications Interfaces

QuickHire relies on internet-based communication technologies to allow interaction between users, servers, and external services.

3.4.1 Web Communication Protocols

The system shall use standard internet protocols including:

- **HTTP (Hypertext Transfer Protocol):** A standard internet communication protocol used by the QuickHire system to enable data transmission between the user's web browser and the QuickHire server. It serves as the foundation for how information is requested and delivered across the web, allowing users to interact with the platform through browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.
- **HTTPS (Secure Hypertext Transfer Protocol):** The secure version of HTTP used by QuickHire to ensure that all data transmitted between the user's browser and the QuickHire server is encrypted and protected. The SRS specifically requires that HTTPS be used as the primary communication protocol to safeguard sensitive user information such as login credentials, profile data, and job application records. It works alongside other security measures in the system such as secure login authentication and password encryption to protect user privacy and comply with data security regulations.

HTTPS shall be used to ensure **secure data transmission** between the user's browser and the QuickHire server.

3.4.2 Email Communication

The system shall send automated emails to users for events such as:

- **Account creation confirmation:** An automated email triggered and sent to a user immediately after they successfully complete the registration process on the QuickHire platform. It serves as a verification that the user's account has been created and is ready to use, allowing them to proceed with setting up their profile and accessing the system's features such as job searching and interview scheduling.
- **Job application submission:** An automated email notification sent to a job seeker after they have successfully submitted a job application through the QuickHire platform. It confirms that their application has been received and recorded in the system's database, and may contain a link directing the user back to the QuickHire platform to monitor their application status.
- **Interview scheduling:** An automated email notification sent to both job seekers and employers when an interview has been arranged through the QuickHire platform. It informs the relevant parties of the scheduled online interview session, which may be conducted through an integrated third-party video communication service, and may contain a link back to the platform for reference.
- **System notifications:** Automated emails generated by the QuickHire platform to inform users of any other important platform-related events or updates that do not fall under the specific categories above. These may also appear as real-time dashboard alerts within the platform, such as incoming interview call alerts or profile visibility updates, keeping users informed of relevant activity on their accounts.

Emails may contain links directing users back to the QuickHire platform.

3.4.3 Data Security

To protect user information, the system shall implement:

- **Secure login authentication:** A security measure implemented by the QuickHire system to verify and validate the identity of users before granting them access to the platform. It ensures that only registered and authorized users, whether job seekers or employers, are able to log in and access their respective accounts and features. This is part of the system's overall commitment to complying with applicable data privacy and security regulations to protect user information.
- **Password encryption:** A security mechanism applied by the QuickHire system to protect user passwords by converting them into an unreadable format before storing them in the database. This ensures that even if the database is compromised, user passwords remain protected and cannot be easily accessed or misused by unauthorized parties. It works alongside secure login authentication as part of the platform's layered approach to data security.
- **Encrypted communication using HTTPS:** A security requirement mandated by the QuickHire SRS that ensures all data transmitted between the user's web browser and the QuickHire server is encoded and protected from interception or tampering. By using HTTPS as the standard communication protocol, the system safeguards sensitive information exchanged during user interactions, such as login credentials, profile updates, job application records, and interview schedules, from being accessed by unauthorized third parties.

3.4.4 Data Synchronization

The system shall ensure that:

- **Profile updates are immediately reflected in the system:** This means that whenever a job seeker or employer makes changes to their profile information through the Profile Editing Interface such as updating their desired job role, hourly rate, skills, educational background, employment type, or uploading a new resume or profile photo, those changes are instantly saved and displayed across the platform without any delay. This ensures that employers viewing a job seeker's profile always see the most current and accurate information, supporting the platform's goal of efficient and up-to-date job matching.

- **Job application status updates are synchronized with the database:** This means that whenever there is a change in the status of a job seeker's application such as it being reviewed, accepted, or rejected by an employer that change is immediately recorded and updated in the QuickHire database in real time. This ensures consistency and accuracy of application records across the system, so that both job seekers and employers always see the same and most current application status, and that automated email notifications reflecting those status changes are triggered accurately and promptly.
- **Interview notifications are delivered in real time:** This means that whenever an interview is scheduled, modified, or initiated by an employer through the QuickHire platform, the corresponding notification is instantly delivered to the relevant job seeker without any delay. This may appear as a dashboard alert within the platform, such as an incoming interview call notification, or as an automated email containing interview details. This real-time delivery is critical to ensuring that job seekers are promptly informed and can respond or prepare accordingly, supporting the platform's goal of efficient communication between employers and applicants.

Reliable internet connectivity is required for communication between the user and the QuickHire system.

4. System Features

4.1 User Registration and Login

4.1.1 Description and Priority

This feature serves as the entry point for the QuickHire platform. It allows job seekers and employers to create accounts and securely log in to access all other system features and functionalities.

Priority: **High**

4.1.2 Stimulus/Response Sequences

Registration:

- **Stimulus:** A new user navigates to the QuickHire registration page and submits their account details.
- **Response:** The system shall validate the input, create the account, and send an automated account creation confirmation email to the user.

Login:

- **Stimulus:** A registered user enters their credentials on the login page.
- **Response:** The system shall authenticate the user through secure login authentication and redirect them to their dashboard.

Invalid Login:

- **Stimulus:** A user enters incorrect login credentials.
- **Response:** The system shall display an error message indicating invalid input and prompt the user to try again.

Logout:

- **Stimulus:** A user clicks the Logout option in the navigation sidebar.
- **Response:** The system shall end the active session and redirect the user to the login page.

4.1.3 Functional Requirements

- **[FR-01]:** The system shall allow job seekers and employers to register by providing the required account information.
 - **[FR-02]:** The system shall send an automated account creation confirmation email upon successful registration.
 - **[FR-03]:** The system shall authenticate users through secure login authentication before granting access to the platform.
 - **[FR-04]:** The system shall encrypt user passwords before storing them in the database.
 - **[FR-05]:** The system shall display an error message when invalid or incorrect login credentials are entered.
 - **[FR-06]:** The system shall allow users to log out through the navigation sidebar, ending their active session.
-

4.2 Profile Management

4.2.1 Description and Priority

This feature allows job seekers to create and manage their professional profiles, including uploading a resume, selecting skills, and updating personal and professional information to be visible to employers.

Priority: **High**

4.2.2 Stimulus/Response Sequences

Profile Editing:

- **Stimulus:** A user clicks the Edit Profile button on the dashboard or navigation sidebar.
- **Response:** The system shall display the profile editing page with all editable fields populated with the user's current information.

Saving Profile Updates:

- **Stimulus:** A user updates one or more profile fields and clicks Save Profile.
- **Response:** The system shall validate the input, save the changes, display a confirmation message, and immediately reflect the updates across the platform.

File Upload:

- **Stimulus:** A user uploads a new profile photo or resume file in PDF format.
- **Response:** The system shall accept the uploaded file and update the profile accordingly.

Skills Selection:

- **Stimulus:** A user navigates to the skills selection panel, searches for a skill, and selects it via checkbox.
- **Response:** The system shall add the selected skill to the user's profile summary.

Invalid Input:

- **Stimulus:** A user enters invalid data in a profile field.
- **Response:** The system shall display an error message indicating the invalid input and prompt the user to correct it.

4.2.3 Functional Requirements

- **[FR-07]:** The system shall provide a profile editing page with labeled fields including job role, hourly rate, available working hours, country, employment type, English proficiency, educational background, portfolio link, age, gender, profile description, and resume upload.
 - **[FR-08]:** The system shall allow users to upload or replace their profile photo and resume file in PDF format.
 - **[FR-09]:** The system shall provide a skills selection panel organized into categories such as Artificial Intelligence, Analytics, API, Backend, Business, Cloud, Database, Design, and Development.
 - **[FR-10]:** The system shall allow users to search for skills using a search bar and select multiple skills through checkboxes.
 - **[FR-11]:** The system shall immediately reflect all profile updates across the platform upon saving.
 - **[FR-12]:** The system shall display a confirmation message after a successful profile update.
 - **[FR-13]:** The system shall display an error message when invalid input is detected in any profile field.
-

4.3 Job Search and Employer Discovery

4.3.1 Description and Priority

This feature allows job seekers to search for and browse available employers and job opportunities on the QuickHire platform. It is the core function that directly fulfills the platform's goal of helping IT professionals find suitable employment quickly and efficiently.

Priority: **High**

4.3.2 Stimulus/Response Sequences

Initiating Job Search:

- **Stimulus:** A user clicks the Find Employer button on the dashboard or navigation sidebar.
- **Response:** The system shall display a list of available employers or job opportunities relevant to the user's profile.

Applying Filters:

- **Stimulus:** A user applies filters to narrow down the job search results.
- **Response:** The system shall update the displayed results based on the selected filter criteria.

Viewing a Job Listing:

- **Stimulus:** A user selects a specific employer or job listing from the search results.
- **Response:** The system shall display detailed information about the selected employer or job opportunity.

No Results Found:

- **Stimulus:** No job listings match the user's search criteria.
- **Response:** The system shall display a message indicating that no results were found and suggest that the user adjust their search criteria.

4.3.3 Functional Requirements

- **[FR-14]:** The system shall provide a job search and employer discovery feature accessible from both the dashboard and the navigation sidebar.
- **[FR-15]:** The system shall allow job seekers to search and filter job opportunities based on relevant criteria.
- **[FR-16]:** The system shall display detailed employer or job listing information when selected by the user.
- **[FR-17]:** The system shall display an appropriate message when no results match the user's search criteria.
- **[FR-18]:** The system shall ensure that all job listings and employer profiles displayed to job seekers are current and synchronized with the database.

4.4 Interview Scheduling and Notification

4.4.1 Description and Priority

This feature enables employers to schedule interviews with job seekers and ensures that both parties are notified in real time. It directly supports the hiring process by facilitating efficient communication and online interview management between employers and applicants.

Priority: **High**

4.4.2 Stimulus/Response Sequences

Scheduling an Interview:

- **Stimulus:** An employer selects a job seeker and schedules an interview through the platform.
- **Response:** The system shall record the interview schedule in the database and send a real-time interview invitation notification to the job seeker.

Receiving an Interview Notification:

- **Stimulus:** A job seeker receives an interview invitation from an employer.
- **Response:** The system shall display an incoming interview call alert on the job seeker's dashboard and send an automated interview scheduling email containing relevant details and a link back to the platform.

Modifying or Cancelling an Interview:

- **Stimulus:** An employer modifies or cancels a scheduled interview.
- **Response:** The system shall update the interview record in the database and deliver a real-time notification to the affected job seeker.

Conducting an Online Interview:

- **Stimulus:** Both employer and job seeker join the scheduled interview session.
- **Response:** The system shall facilitate the online interview through an integrated third-party video communication service.

4.4.3 Functional Requirements

- **[FR-19]:** The system shall allow employers to schedule interviews with job seekers through the platform.
- **[FR-20]:** The system shall store all interview schedules in the database and synchronize them in real time.
- **[FR-21]:** The system shall deliver real-time interview notifications to job seekers upon scheduling, modification, or cancellation of an interview.
- **[FR-22]:** The system shall send automated interview scheduling emails to both employers and job seekers containing relevant details and a link back to the platform.
- **[FR-23]:** The system shall display incoming interview call alerts on the job seeker's dashboard.
- **[FR-24]:** The system shall support integration with third-party video communication services to facilitate online interviews between employers and job seekers.

4.5 Feedback and Reporting

4.5.1 Description and Priority

This feature allows both job seekers and employers to post feedback and report users on the QuickHire platform. It supports platform integrity and user trust by providing a mechanism for users to evaluate interactions and flag inappropriate behavior.

Priority: **Medium**

4.5.2 Stimulus/Response Sequences

Posting Feedback:

- **Stimulus:** A user navigates to the feedback section and submits a review or rating for another user.
- **Response:** The system shall record the feedback in the database and display a confirmation message to the user.

Reporting a User:

- **Stimulus:** A user selects the option to report another user for inappropriate behavior or content.
- **Response:** The system shall record the report in the database and flag it for administrative review.

Invalid Submission:

- **Stimulus:** A user submits feedback or a report with incomplete or invalid information.
- **Response:** The system shall display an error message prompting the user to correct the input before resubmitting.

4.5.3 Functional Requirements

- **[FR-25]:** The system shall allow job seekers and employers to post feedback about their interactions on the platform.
- **[FR-26]:** The system shall allow users to report other users for inappropriate behavior or content.
- **[FR-27]:** The system shall display a confirmation message after feedback or a report has been successfully submitted.
- **[FR-28]:** The system shall display an error message when feedback or report submissions contain invalid or incomplete information.
- **[FR-29]:** The system shall record all submitted feedback and reports in the database for administrative review.

4.6 Job Listing Management

4.6.1 Description and Priority

This feature allows employers to create, manage, and publish job listings on the QuickHire platform. It enables employers to define job requirements and allows job seekers to view and apply for available positions.

Priority: **High**

4.6.2 Stimulus/Response Sequences

Creating a Job Listing:

- **Stimulus:** An employer navigates to the job listing section and submits job details.

- **Response:** The system shall validate the input, save the job listing, and make it visible to job seekers.

Editing a Job Listing:

- **Stimulus:** An employer selects an existing job listing and updates its details.
- **Response:** The system shall update the listing and reflect the changes immediately.

Deleting a Job Listing:

- **Stimulus:** An employer chooses to delete a job listing.
- **Response:** The system shall remove the job listing from the database and platform.

Viewing Job Listings:

- **Stimulus:** A job seeker browses available job listings.
- **Response:** The system shall display all active job listings with relevant details.

4.6.3 Functional Requirements

[FR-30]: The system shall allow employers to create job listings by providing job title, description, required skills, employment type, and compensation details.

[FR-31]: The system shall allow employers to edit and update existing job listings.

[FR-32]: The system shall allow employers to delete job listings from the platform.

[FR-33]: The system shall display all active job listings to job seekers in the job search interface.

[FR-34]: The system shall ensure that all job listings are stored in and synchronized with the database in real time.

4.7 Messaging and Communication

4.7.1 Description and Priority

This feature enables direct communication between job seekers and employers through an in-platform messaging system. It supports efficient coordination, clarification, and interaction before interviews.

Priority: High

4.7.2 Stimulus/Response Sequences

Sending a Message:

- **Stimulus:** A user composes and sends a message to another user.
- **Response:** The system shall deliver the message in real time and store it in the database.

Receiving a Message:

- **Stimulus:** A user receives a message from another user.
- **Response:** The system shall display the message in the chat interface and notify the user.

Viewing Conversation:

- **Stimulus:** A user opens a conversation thread.
- **Response:** The system shall display the full message history between the users.

Invalid Message:

- **Stimulus:** A user attempts to send an empty or invalid message.
- **Response:** The system shall display an error message and prevent sending.

4.7.3 Functional Requirements

[FR-35]: The system shall allow job seekers and employers to send and receive messages within the platform.

[FR-36]: The system shall store all messages in the database and maintain conversation history.

[FR-37]: The system shall deliver messages in real time or near real time.

[FR-38]: The system shall notify users when a new message is received.

[FR-39]: The system shall display an error message when a user attempts to send an empty or invalid message.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **[NFR-01] Response Time:** The system shall load all pages and complete user-initiated actions such as profile updates, job searches, and application submissions within an acceptable response time to ensure a seamless user experience across all supported devices and browsers.
- **[NFR-02] Real-time Notification Delivery:** The system shall deliver interview notifications and application status updates to users in real time, with no significant delay between the triggering event and the notification appearing on the user's dashboard or inbox.
- **[NFR-03] Data Synchronization Speed:** The system shall immediately synchronize all profile updates, job application status changes, and interview schedule modifications with the database upon submission, ensuring consistency across all active user sessions.
- **[NFR-04] Concurrent User Handling:** The system shall be capable of supporting multiple simultaneous users, including job seekers and employers, performing actions such as profile editing, job searching, and interview scheduling without application failure or significant performance degradation.
- **[NFR-05] File Upload Performance:** The system shall process and store uploaded files such as profile photos and PDF resumes within an acceptable time frame, ensuring that users are not required to wait excessively during the upload process.

5.2 Safety Requirements

- **[NFR-06] Fail-Safe Notification Delivery:** In the event of a system error or notification service failure, the system shall default to delivering all critical communications such as interview invitations and application status updates through the integrated email service to prevent loss of important information.
- **[NFR-07] Data Integrity on Profile Updates:** To prevent data corruption, the system shall use reliable write operations when saving profile updates, resume uploads, and application records to the database, ensuring that no partial or incomplete data is stored.
- **[NFR-08] Session Safety:** The system shall automatically terminate inactive user sessions after a defined period of inactivity to prevent unauthorized access to user accounts, particularly on shared or public devices.

5.3 Security Requirements

- **[NFR-09] Encrypted Communication:** All data transmitted between the user's browser and the QuickHire server shall be protected using HTTPS to prevent unauthorized interception or tampering of sensitive user information.
- **[NFR-10] Password Encryption:** All user passwords shall be encrypted before being stored in the database, ensuring that credentials remain protected even in the event of a database breach.
- **[NFR-11] Secure Authentication:** The system shall implement secure login authentication to verify the identity of all users before granting access to the platform, preventing unauthorized access to job seeker and employer accounts.
- **[NFR-12] Data Privacy Compliance:** The system shall comply with applicable data privacy and security regulations to protect all user information stored and processed within the QuickHire platform, including personal details, resume files, and application records.
- **[NFR-13] Permission-Based Access Control:** The system shall enforce role-based access control, ensuring that job seekers and employers can only access features and data relevant to their respective user classes, preventing unauthorized access to privileged functions.

5.4 Software Quality Attributes

- **[NFR-14] Availability:** The QuickHire platform shall maintain high availability to ensure that job seekers and employers can access the system at any time, minimizing downtime that could disrupt the job search and hiring process.
- **[NFR-15] Usability:** The user interface shall follow modern web design standards and remain simple, responsive, and easy to navigate, supporting users with basic to moderate technical skills across all supported devices including desktops, tablets, and smartphones.
- **[NFR-16] Compatibility:** The system shall be fully compatible with modern web browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari, and shall support common operating systems including Windows, macOS, Android, and iOS.
- **[NFR-17] Maintainability:** The system shall follow standard coding and design conventions to ensure that future updates and maintenance can be performed efficiently, even if handled by a different development team.
- **[NFR-18] Scalability:** The system shall be built on standard cloud hosting services and database platforms to support growth in the number of users, job listings, and application records without significant performance degradation.

5.5 Business Rules

- **[NFR-19] Role-Based Feature Access:** Job seekers and employers shall only be granted access to features appropriate to their user class. Job seekers shall have access to profile management, job search, and interview participation, while employers shall have access to applicant evaluation, job management, and interview scheduling features.
- **[NFR-20] Profile Visibility Control:** A job seeker's profile shall only be visible to employers when the job seeker's profile status is set to active or visible, giving users control over their own discoverability on the platform.
- **[NFR-21] Resume File Format Restriction:** The system shall only accept resume uploads in PDF format to ensure consistency and compatibility across all employer devices and browsers.
- **[NFR-22] Scope Limitation:** The QuickHire platform shall strictly limit its functionalities to the job discovery, application, and interview processes. Advanced human resource management functions such as payroll processing, employee performance tracking, and post-employment management shall not be included within the scope of the system.
- **[NFR-23] Third-Party Service Dependency Disclosure:** The system shall inform users that certain features such as email notifications and online interviews rely on third-party services, and that changes to those services may affect platform functionality.

6. Other Requirements

6.1 Database Requirements

[FR-40] Centralized Data Storage:

The system shall utilize a centralized database management system to store and manage all system data, including user accounts, profiles, job listings, applications, messages, interview schedules, and notifications.

[FR-41] Schema Integrity:

The database schema shall include structured and normalized tables such as *Users*, *Profiles*, *Job_Listings*, *Job_Applications*, *Skills*, *Resumes*, *Interview_Schedules*, *Messages*, and *Notifications* to ensure efficient data retrieval and consistency.

[FR-42] Real-Time Data Synchronization:

The system shall ensure that updates to job listings, messages, applications, and profiles are immediately synchronized with the database and reflected across the platform.

[FR-43] Data Relationship Management:

The system shall maintain proper relationships between entities (e.g., Users–Profiles, Employers–Job_Listings, Job_Seekers–Applications, Users–Messages) to support accurate data linking and retrieval.

[FR-44] Message Storage and Retrieval:

The system shall store all user messages securely and allow retrieval of complete conversation history between job seekers and employers.

Appendix A: Glossary

This section defines the technical and domain-specific terminology essential for the development and use of the QuickHire system.

Authentication:

The process of verifying the identity of a user (job seeker or employer) before granting access to the system through login credentials.

Database Management System (DBMS):

A software system used to store, manage, and retrieve structured data such as user profiles, job listings, applications, messages, and interview schedules.

Encryption:

The process of converting sensitive data (e.g., passwords and messages) into a secure format to prevent unauthorized access.

HTTPS (Hypertext Transfer Protocol Secure):

A secure communication protocol used to encrypt data transmitted between the user's browser and the QuickHire server.

Job Listing:

A post created by an employer that contains details about a job opportunity, including job title, description, required skills, employment type, and compensation.

Job Application:

A record of a job seeker's submission to a job listing, including relevant profile and resume information.

Job Seeker:

A primary user of the system who creates a profile, searches for job opportunities, sends messages, and participates in interviews.

Employer / Recruiter:

A user who creates job listings, reviews applicants, communicates with job seekers, and schedules interviews.

Messaging System:

An in-platform communication feature that allows job seekers and employers to exchange messages in real time.

Conversation Thread:

A collection of messages exchanged between two users, displayed in chronological order within the messaging interface.

Profile Management:

The process of creating, updating, and maintaining user information such as skills, resume, and job preferences.

Resume (Curriculum Vitae):

A document uploaded by job seekers that contains their qualifications, experience, and skills for employer evaluation.

Real-Time Notification:

System-generated alerts (e.g., interview invitations, messages, or application updates) delivered instantly to users via dashboard or email.

Role-Based Access Control (RBAC):

A security mechanism that restricts system access based on user roles (e.g., job seeker or employer).

Scalability:

The ability of the system to handle increasing numbers of users, job listings, messages, and data without performance degradation.

Third-Party Integration:

The use of external services (e.g., email providers or video conferencing tools) to extend system functionality.

User Interface (UI):

The visual components of the system (e.g., dashboard, forms, job listings, chat interface) that allow users to interact with QuickHire.

Cloud Hosting:

The use of remote servers hosted on the internet to store, manage, and process system data instead of local servers.

Appendix B: Analysis Models

This section presents simplified analysis models that describe the structure and behavior of the QuickHire system.

Appendix C: To Be Determined List

This section identifies items that are not yet fully defined and may require further clarification in future revisions.

- **[TBD-01] Video Interview Platform:**
The specific third-party video conferencing tool to be integrated (e.g., Zoom, Google Meet) is yet to be finalized.
- **[TBD-02] Advanced Job Matching Algorithm:**
The possible use of AI or recommendation algorithms for improved job matching is still under consideration.
- **[TBD-03] Notification Channels Expansion:**
Support for additional notification channels (e.g., SMS or mobile push notifications) may be added in future versions.
- **[TBD-04] Admin Dashboard Features:**
Detailed administrative controls for managing users, reports, and system monitoring are not yet fully specified.
- **[TBD-05] Payment Integration:**
If premium features or paid job postings are introduced, the payment gateway and process will need to be defined.